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Winpack Microsoft Intune – Tata Advanced Systems Limited (TASL)

A PROJECT REPORT

Submitted by

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "MICROSOFT INTUNE IN SECURING DEVICES: BALANCING SECURITY FEATURES AND USER EXPERIENCE IN ENTERPRISE ENVIRONMENTS" is a bonafide work of "Syed Saifulla H (20231CCS3004) who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY during 2026-27.

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DECLARATION

Iam the student of final year B.Tech in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY at Presidency University, Bengaluru, named Syed Saifulla H, hereby declare that the project work titled **"MICROSOFT INTUNE IN SECURING DEVICES: BALANCING SECURITY FEATURES AND USER EXPERIENCE IN ENTERPRISE ENVIRONMENTS"** who has successfully carried out the project work at **Tata Advanced Systems Limited (TASL)** and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (CYBER SECURITY) during the academic year of 2026-27. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other Institution.

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Abstract

The proliferation of cloud computing and mobile technologies has fundamentally transformed the modern enterprise landscape, enabling unprecedented levels of workforce flexibility and operational efficiency. However, this digital transformation has simultaneously introduced significant security vulnerabilities, as organizations struggle to protect sensitive corporate data across an increasingly diverse and distributed device ecosystem. The challenge is further compounded by the need to maintain seamless user productivity and satisfaction, creating a critical tension between security enforcement and user experience. Microsoft Intune, a cloud-native enterprise mobility management (EMM) solution, provides organizations with a comprehensive framework for managing and securing devices, applications, and data across multiple platforms including iOS, Android, Windows, and macOS. Despite its widespread adoption, the actual effectiveness of Intune's security features and their real-world impact on end-user experience remains insufficiently quantified in existing literature.

Quantitative analysis revealed statistically significant variations in user impact across different security controls. Device encryption, which operates silently in the background, exhibited the lowest perceived user disruption with a mean impact score of 0.57 on a 0–2 scale. Conversely, multi-factor authentication demonstrated the highest mean impact of 1.57, reflecting the cumulative friction introduced by repeated authentication prompts during daily workflows. Remote wipe functionality showed the greatest variability in impact (standard deviation 1.07), indicating that user perception is heavily influenced by contextual factors such as device ownership and the circumstances under which remote wipe is invoked. Application protection policies yielded the most consistent impact (standard deviation 0.49), suggesting that users experience uniform restrictions across their managed applications.

Beyond the statistical analysis, this research also developed and validated a proof-of-concept integration between Intune and physical access control systems (WinPak) at the participating organization, enabling automated, context-aware policy enforcement based on employee presence. This integration demonstrated that security policies can be dynamically applied and removed in response to real-world physical events, reducing manual IT intervention while maintaining robust security postures.

The findings contribute both academic and practical insights by providing a quantitative trade-off framework that enables security administrators to make evidence-based decisions when configuring Intune policies. The framework allows organizations to prioritize high-impact features only where risk exposure justifies the usability cost, thereby achieving an optimal balance between security effectiveness and user experience. This research underscores that successful enterprise mobility management requires not only technical proficiency but also a deep understanding of human factors and organizational context.

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Abbreviations

Abbreviation	**Full Form**
AEHMS	Automated Equipment Health Monitoring System
AFV	Armoured Fighting Vehicle
API	Application Programming Interface
ATGM	Anti Tank Guided Missile
AUC-ROC	Area Under Curve - Receiver Operating Characteristic
BMCS	Bi-Modular Charge System
COTS	Commercial Off-The-Shelf
CSV	Comma-Separated Values
DHT	Digital Humidity and Temperature
DRDO	Defence Research and Development Organisation
EME	Electronics and Mechanical Engineers
ERP	Enterprise Resource Planning
FPA	Focal Plane Array
GPIO	General Purpose Input/Output
GPS	Global Positioning System
HAA	High Altitude Area
HTML	HyperText Markup Language
IFF	Identification of Friend or Foe
InfluxDB	Time-Series Database
IoT	Internet of Things
IRNSS	Indian Regional Navigation Satellite System
JSON	JavaScript Object Notation
LAC	Line of Actual Control
LTE	Long Term Evolution
MAWS	Missile Approach Warning System
MCCS	Mobile Cellular Communication System

MGB	Main Gear Box
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
MTBF	Mean Time Between Failures
NVD	Night Vision Device
OEM	Original Equipment Manufacturer
PESTEL	Political, Economic, Social, Technological, Environmental, Legal
QoS	Quality of Service
RBAC	Role-Based Access Control
REST	Representational State Transfer
RF	Random Forest
RH	Relative Humidity
ROI	Return on Investment
SAM	Surface to Air Missile
SAR	Synthetic Aperture Radar
SDG	Sustainable Development Goal
SDK	Software Development Kit
SQLite	Structured Query Language Lite
SSL	Secure Sockets Layer
SVM	Support Vector Machine
TBA	Tactical Battle Area
TI	Thermal Imaging
TLS	Transport Layer Security
UAV	Unmanned Aerial Vehicle
URL	Uniform Resource Locator
VPN	Virtual Private Network
XML	Extensible Markup Language